

Replication Report: Clamped Pockels Tensor of Tetragonal BaTiO₃ de Mestral et al. 2025 (arXiv:2506.13209)

Automated replication (Hermes subagent)

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1 Target and headline

The paper computes the *clamped* Pockels tensor of tetragonal BaTiO₃ (BTO) from DFT (Quantum ESPRESSO), the modern theory of polarization, and finite-difference electro-optic derivatives (AiiDA-Vibroscope). The technological headline is the largest Pockels coefficient r_{51} , experimentally 730 ± 150 pm/V when clamped.

Corpus mislabel. The directory is named `textures-spin-mestral2025`, but the paper contains *no spin physics*; it is a ferroelectric electro-optics study. The provided `gobel2024` skyrmion Kubo kernel and `spin.ed_probes` were inspected and found irrelevant, so they were not used. We replicated the paper’s actual physics.

2 Physics replicated

Per Eq. (4) of the paper, the ionic Pockels response is a sum over Γ -point optical modes,

$$r_{ijk} = -\frac{1}{n_i^2 n_j^2} \chi_{ijk}^{(2)} - \frac{1}{n_i^2 n_j^2} \sum_m \frac{\alpha_{ij}^m p_{m,k}}{\omega_m^2}, \quad (1)$$

with α^m the Raman susceptibility, $p_{m,k}$ the mode polarity, and ω_m the mode frequency. Because the in-plane *soft* optical mode has the smallest ω_m , the $1/\omega_m^2$ factor makes r_{51} soft-mode dominated: as Ti off-centering is reduced, the soft mode softens and r_{51} diverges.

Model 1 (mode-sum, $1/\omega^2$). We calibrate $K = \alpha p/(n_i n_j)$ once at the P4bm ground state (0.466%, $r_{51} = 391.2$ pm/V, $\omega = 2.0$ THz) and *predict* the rest of the Table IV series from the paper’s ω values alone.

Model 2 (Landau double-well soft mode). Free energy $F(Q) = \frac{1}{2}aQ^2 + \frac{1}{4}bQ^4$ with $a < 0, b > 0$ gives $\omega^2(Q) \propto 3q^2 - 1$ ($q = Q/Q_0$), reproducing the softening trend as off-centering Q decreases below the ground state.

3 Results

At 0.45% displacement the model gives $r_{51} = 695.5$ pm/V, inside the experimental band 730 ± 150 pm/V, reproducing the paper’s central claim. The crude Landau $\omega(\text{disp})$ model is qualitatively correct (softening as off-centering drops) but quantitatively poor (MARE $\approx 33\%$): the true ω collapse is far steeper than a simple quartic double-well, reflecting strong anharmonicity near the transition.

Ti disp	paper r_{51}	model r_{51}	rel. err.
0.466% (anchor)	391.2	391.2	0.0%
0.450%	667.0	695.5	+4.3%
0.425%	1614.7	1564.8	-3.1%

Table 1: r_{51} (pm/V): mode-sum model vs. paper Table IV. Two of three points are genuine predictions from a single anchor. $\text{MARE}(\text{predicted}) = 3.68\%$.

4 Verdict

PARTIAL $\rightarrow$ leaning REPLICATED for the headline. The r_{51} magnitude, its $1/\omega^2$ soft-mode dominance, and the experimental bracketing at $\sim 0.45\%$ are all reproduced to $< 5\%$. The microscopic ω (off-centering) law is only qualitatively captured. Coverage 7/10, Agreement 8/10.

5 Credits

DFT reference data: de Mestral et al. 2025. Skyrmion Kubo kernel (`gobel2024_sd_skyrmion_kubo_Lz_kernel.py`) inspected but not used (physically irrelevant to this paper).